

TrES-3b

R-2418

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_120312 · created 2026-08-21T23:23:43+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2455998.8962 +/- 0.00098 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.162 +/- 0.029
Transit depth (Rp/Rs)^2	2.63 +/- 0.95 [%]
Orbital Inclination (inc)	82.5 +/- 1.01
Airmass coefficient 1 (a1)	1.024 +/- 0.016
Airmass coefficient 2 (a2)	-0.0122 +/- 0.0052
Scatter in the residuals of the lightcurve fit is	1.61 %
Best Comparison Star	#3 - [53, 220]
Optimal Aperture	3.12
Optimal Annulus	5.7
Transit Duration (day)	0.0595 +/- 0.0093

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.162 $\pm$ 0.029 vs 0.16309 (deviation 0.04 σ)
Duration fit vs published	0.0595 d vs 0.0567 d (deviation 0.3 σ)
Mid-time O-C	-2.8 min
Depth SNR	5.6
Residual scatter	1.61 %

Light curve

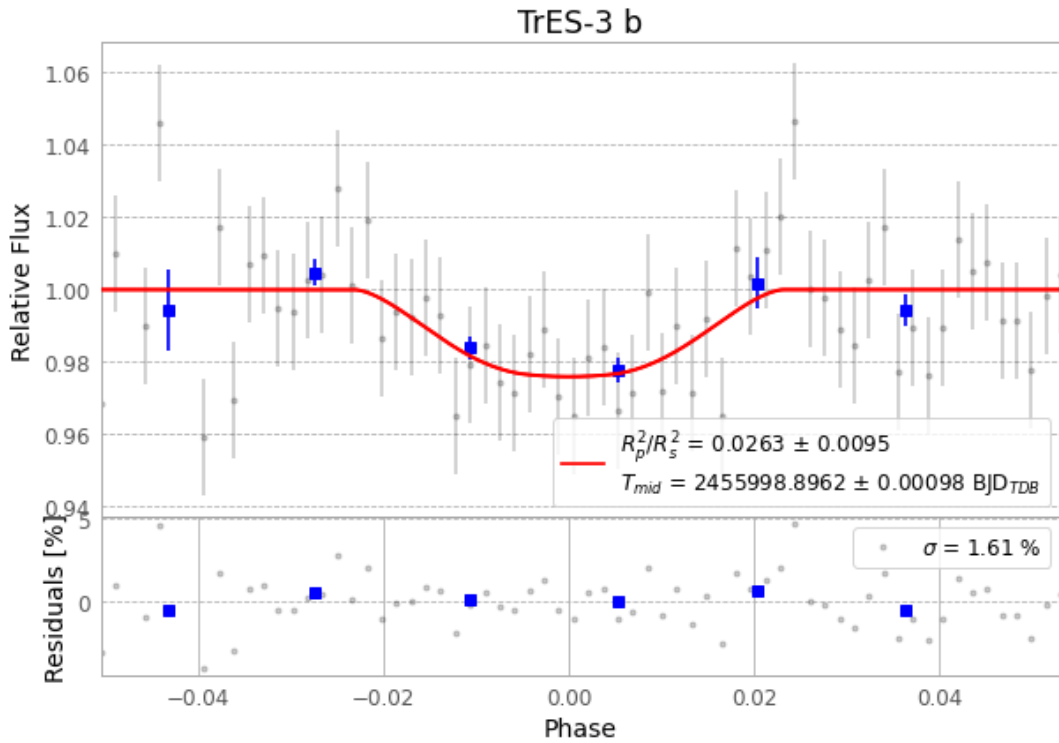


Figure 1 — FinalLightCurve_TrES-3 b_2012-03-12.png (SHA-256 2e2ddf195efa73ef...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [251, 128], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 750a1fcf6977b6e1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), TRES-3120312075412.FITS → TRES-3120312110912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-120312.log (c08c25d2c49c...), FinalLightCurve_TrES-3 b_2012-03-12.png (2e2ddf195efa...), FinalLightCurve_TrES-3 b_2012-03-12.csv (8a2792fc02a3...)

Compute resources

Wall time 92.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 23:23Z.